

SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Milestone 1 Report

Intern: Mohammed Ateeq Ur Rahman

Program: Infosys Virtual Internship

Project: SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Framework: YOLOv8 (Ultralytics)

Hardware Used: NVIDIA GTX 1650 Ti GPU (4GB)

Software: Python, PyTorch, Ultralytics YOLOv8, OpenCV, VS Code

Project Overview

SafetyEye is an AI-powered workplace monitoring system designed to improve safety compliance in construction and industrial environments. The system uses computer vision and deep learning techniques to automatically detect safety-related objects such as helmets, safety vests, workers, and machinery.

The primary goal of this system is to monitor workplace environments and identify potential safety violations, thereby improving worker safety and reducing risks.

Milestone 1 Objectives

The main objectives of Milestone 1 were:

- Setting up the development environment
 - Exploring and verifying the dataset
 - Preparing the dataset for training
 - Configuring the YOLOv8 training pipeline
 - Performing initial model training
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Week 1 – Environment Setup and Dataset Exploration

Objective

The objective of Week 1 was to set up the development environment, understand the dataset structure, and verify dataset quality to ensure smooth model training.

Environment Setup

The development environment was configured using the following tools and technologies:

- Python
- Virtual Environment (venv)
- PyTorch
- Ultralytics YOLOv8
- OpenCV
- NumPy
- Matplotlib
- VS Code

A virtual environment was created to manage dependencies efficiently and avoid conflicts between packages.

GitHub Repository Setup

The project repository was cloned from GitHub and organized into a structured format. The project directories include:

- dataset (images and labels)
- scripts
- docs
- notebooks

This structure ensures proper organization and easier maintenance of the project.

Dataset Exploration

The dataset used in this project is the **Construction Site Safety Dataset**, obtained from Roboflow.

Dataset Details

- Total Images: 2801
- Annotation Format: YOLOv8
- Number of Classes: 10

Classes in Dataset

- Hardhat
- Mask
- NO-Hardhat
- NO-Mask
- No-Safety Vest
- Person
- Safety Cone
- Safety Vest
- Machinery
- Vehicle

Dataset Verification

Custom scripts were developed to verify the dataset quality. The verification process included checking for:

- Missing label files
- Empty label files
- Incorrect label formats

Verification Results

- Training: 1960 images, 0 missing labels, 15 empty labels
- Validation: 560 images, 0 missing labels, 7 empty labels
- Test: 281 images, 0 missing labels, 2 empty labels

The dataset was successfully verified and confirmed to be suitable for training.

Image Visualization

OpenCV was used to visualize sample images to ensure that:

- Images are readable
 - Labels match the objects correctly
 - Bounding boxes are properly aligned
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Week 2 – Dataset Preparation and Initial Training

Objective

The objective of Week 2 was to prepare the dataset and verify the YOLO training pipeline.

Dataset Splitting

The dataset was divided into three parts:

- Training Set: 70% (1960 images)
- Validation Set: 20% (560 images)
- Testing Set: 10% (281 images)

This split ensures proper model training and evaluation.

Class Identification

A script was used to scan label files and identify class IDs.

Detected class IDs: 0 to 9

Total classes detected: 10

These class IDs were mapped to their corresponding class names in the data.yaml file.

YOLO Configuration

The dataset was configured using a YOLOv8 configuration file (data.yaml), which included:

- Dataset path
 - Train, validation, and test paths
 - Class names
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Initial Model Training

A preliminary training run was performed to verify the YOLO training pipeline.

Training configuration:

- Model: YOLOv8n
 - Epochs: 1
 - Image Size: 640
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Initial Training Results

- Precision: 0.464
- Recall: 0.330
- mAP@0.5: 0.314
- mAP@0.5:0.95: 0.170

These results confirmed that the training pipeline was working correctly.

Milestone 1 Results

Key Outcomes

- Development environment successfully configured
 - Dataset verified and cleaned
 - Dataset split into training, validation, and testing sets
 - YOLOv8 configuration file created
 - Initial model training pipeline verified
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Dataset Summary

- Training Images: 1960
 - Validation Images: 560
 - Test Images: 281
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Initial Model Performance

- Precision: 0.464
 - Recall: 0.330
 - mAP@0.5: 0.314
 - mAP@0.5:0.95: 0.170
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Milestone 1 Conclusion

Milestone 1 successfully established the foundation for the SafetyEye system. The development environment was configured, the dataset was explored and verified, and the training pipeline was successfully implemented.

The initial YOLOv8 training confirmed that the system is ready for further model training, optimization, and performance improvements in the upcoming milestones.